$\dot{m} = 510 \text{ kg/s}$ $\qquad T_2/T_1 = (P_2/P_1)^{\frac{\gamma-1}{\gamma \eta_{pc}}}$

$r_p = 18$

$$T_2 = 300\,(18)^{0.286/0.82}$$

$$T_2 = 784.5 \text{ K}$$

$$\dot{m} = \rho A V = \frac{P}{RT} \cdot \pi D h \cdot V_a$$

$$510 = \frac{1800}{0.287 \times 784.5} \times \pi \times 2 \times 0.07 \times V_a$$

$$V_a = 202.35 \text{ m/s}$$

$$\dot{m} = \rho_1 A_1 V_{1a} = \rho_2 A_2 V_{2a}$$

$$U = \frac{\pi D N}{60} = \frac{\pi \times 2 \times 3000}{60} = 314.16 \text{ m/s}$$

$$V_a/U = \frac{202.35}{314.16} = 0.644$$

$$\frac{\Delta h_0}{U^2} = \frac{1.0045(784.5 - 300) \times 1000}{314.16^2}$$

بار الکترون

$= 4.931$

بار موثر $= \frac{4.931}{16} = 0.3082$